DrugFindR

High Throughput in silico Screening of the LINCS Small Molecules

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Abstract

The COVID-19 Pandemic has highlighted the importance of being able to rapidly and accurately identify potential drug treatments for emerging diseases. The development of in silico drug repurposing tools is complemented by the increased availability of Next-Generation Sequencing transcriptomic datasets which allows us to identify significant changes in gene expression associated with a given disease state.

Here we introduce DrugFindR, a novel method for rapid screening of small molecules for activity against a given disease transcriptomic signature. Using the DrugFindR Application, users can upload and analyze their own disease transcriptomic signatures and identify candidate small molecules that are discordant or "reverse" the disease signature. DrugFindR utilizes the Library of Integrated Network-based Cellular Signatures (LINCS) as the source of small molecule signatures (¿20,000 small molecule signatures available). Novel candidate drugs are those whose signatures are 1) concordant (similar) to known drug treatments for the disease of interest and 2) are the most likely to reverse the disease signature. This two-pronged approach allows for a more accurate filtering of the candidate drug list. DrugFindR can be applied to rapidly screen candidate drugs in silico for both new and existing disease conditions.

We have applied this method to identify potential treatments for COVID-19 and were able to identify 17 candidates, including 10 with known anti-viral properties and 7 that are currently in clinical trial as therapies for COVID-19.

Introduction

Over the past two decades, the pace of novel drug discovery has slowed down drastically [2]. However, the rising tide of newer diseases has not stopped, as evidenced by the recent COVID-19 pandemic. This has forced scientists to look at repurposing existing drugs for new indications. A particularly attractive outcome is when we can repurpose an existing FDA-approved drug for a different indication as that significantly reduces the amount of evidence needed to establish safety and efficacy and allows for a shorter time to market [5]. There have been several approaches that have been proposed to identify potential novel indications for existing drugs. Many such high-throughput screens rely on bench-work and thus are slower. Several Computatational approaches are also available that rely on complex models to achieve the same [5].

In this poster we propose a simpler technique for high-throughput screening of the LINCS Small molecule database built on a principle of triangulation to quickly identify high-confidence candidate repurposable drugs. We then apply that to the COVID-19 signatures and generate a list of candidate drugs, several of them have known anti-viral properties and are in trial.

Approach

Figure 1 illustrates the principle of triangulation. drugfindR utilizes the existing database of gene perturbation (knockdown and overexpression), and chemical perturbation signatures provided in LINCS as a reference database. It can also take, as input, the disease signature which is then appropriately filtered. It then compares these signatures to generate a *similarity* index. This index is similar to a pearson correlation coefficient and can range from -1 to 1. An ideal candidate drug will have a signature that has negative similarity with the disease signature, positive similarity with the drug signature and appropriate similarity indices with gene knockdown and overexpression signatures.

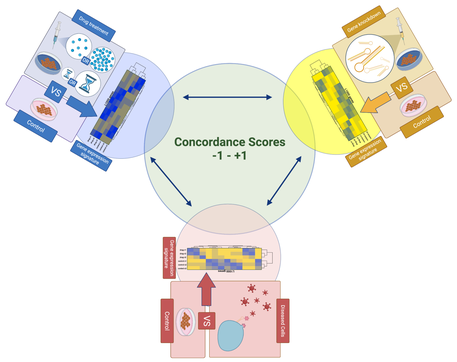

Figure 1: The Triangulation Principle

Application

We used drugfindR to identify a list of potential candidates for treating COVID-19. The COVID-19 disease signatures were obtained from the publicly available RNASeq dataset GSE147507. To provide the second arm for triangulation, we utilized 5 groups of drugs whose signatures were averaged to generate consensus signatures for each drug group. All of these drugs were the ones that were known to have activity against COVID-19 or had strong theoretical basis for the activity. We identified 83 chemical perturbagens with potential activity against COVID-19. After application of several stringent cutoffs and literature review, 20 drugs were designated the *top* candidates. Out of these, 7 drugs were already in clinical trials for treatment of COVID-19. Figure 2 summarizes our results and highlights the significant drugs. The paper detailing this analysis has been recently published [4].

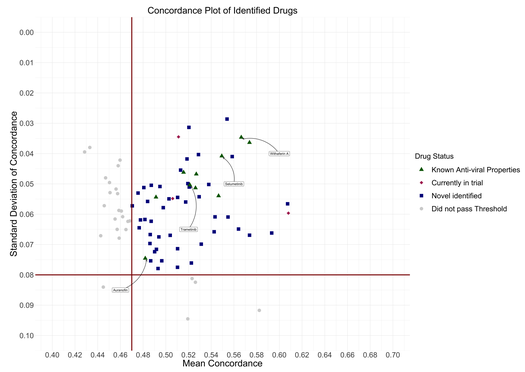

Figure 2: Distribution of Drug Concordances with Significant Drugs Highlighted

Use Cases

The above analysis highlights the use of drugfindR with *fuzzy* drug signatures against disease signatures to identify the drug candidates. In addition, the following use cases have been demonstrated in literature:

- A known drug can be used along with a disease signature to assess the potential of the drug against a disease. The gene knockdowns and overexpression can be used as validation [3].
- A Known Drug-Gene interaction can be used to support a hypothesis of a drug's activity against a particular disease with the disease signature acting as validation [1].

Availability

drugfindR is available as an R Package on github (`https://git.io/J396f`). We are also working on making a shiny-based web application that could be used to generate such analyses quicker.

References

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